

• From Theorem 3, the matrix equation $Ax=0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.

• From the row reduction algorithm, the augmented matrix $[A \ 0]$ can be converted to a echelon form.

• By Theorem 2, since $[A \ 0]$ has a unique solution, each of the columns of A must be pivot columns.

∴ There are exactly 3 pivot columns.

• From condition 1 of the defn of echelon form, each of Row 1 to 3 must have pivots. ∴ Row 4 should be a zero row, i.e. $[0 \ 0 \ 0]$.

• If Row 1 has its pivot in Columns 2 or 3, and taking into consideration Condition 2 from the defn. of echelon form, it will violate the fact that there are exactly 3 pivots.

∴ Row 1 must have its pivot in Column 1, i.e. Row 1 Column 1 = $\square$.

• From condition 3 of the defn of echelon form, Row 2 Column 1 = Row 3 Column 1 = Row 4 Column 1 = 0.

• If Row 2 has its pivot in Column 3, considering condition 2 from the definition of echelon form is True, it will violate the fact that there are exactly 3 pivots.

∴ Row 2 must have its pivot in Column 2 and Row 3 will have its pivot in Column 3, i.e. Row 2 Column 2 = $\square$ and Row 3 Column 3 = $\square$.

• From condition 3 of the defn. of echelon form, Row 3 Column 2 = Row 4 Column 2 = Row 4 Column 3 = 0.

• Since there are no constraints on Row 1 Column 2, Row 1 Column 3 and Row 2 Column 3, each of them will be represented by a $*$.

∴ The echelon form of A is
$$\begin{bmatrix} \square & * & * \\ 0 & \square & * \\ 0 & 0 & \square \\ 0 & 0 & 0 \end{bmatrix} \quad \square$$